

USER MANUAL

Data Integration and Visualization Tool

Version 1.0

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1 Introduction

The Data Integration and Visualization Software is a web-based application designed to manage projects, administrative records, and generate interactive data visualizations from

uploaded datasets. The system allows authorized users to organize files, track project related activities, upload structured data, and create analytical plots for comparative analysis.

This manual provides step-by-step guidance for using the system, including login, project

management, record handling, data upload, visualization, and system settings.

2 User Roles and Access Levels

Access to system features depends on the assigned user role.

Admin

- Create and manage users
- Assign roles and permissions
- View dashboard statistics
- Manage projects
- Access all modules

User

- Access assigned projects
- Upload and manage datasets
- Generate visualizations
- Manage records within allowed modules

2.1 Login Procedure

- Open the application.
- Enter your registered email address.

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- Enter your password.

- Click Login.

If login fails, verify credentials or contact the administrator.

Figure 1: Login Page

3 Admin Dashboard

After successful login, the system opens the Dashboard page. The dashboard provides an overview of system users and allows administrators to create new users.

The User Overview panel displays a summary of all registered users in the system.

The panel includes:

- Total number of users
- Role-wise distribution of users:
 - Group Directors
 - Team Leads
 - System Managers
 - Junior Research Fellows
 - Contract Engineers

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Each role displays the total count along with a visual icon for quick identification.

This

section helps administrators monitor user distribution across different roles.

3.1 User Creation

The "Create New User" section allows administrators to add new users to the system.

To create a new user:

- Enter the User Name.
- Select the appropriate Role from the dropdown list.
- Enter the Login ID (email address).
- Enter the Password.
- Click the Create button.

The system performs validation to ensure all required fields are completed. A confirmation

message is displayed upon successful user creation. The password field includes a visibility

toggle to allow users to view or hide the entered password.

Figure 2: Admin Dashboard Page

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3.2 User Management :

The User Management page allows administrators to view, search, and manage all registered users in the system.

Figure 3: User Management Page

Manage User Login

This section displays a tabular list of all users with the following details:

- Name - Registered user name
- Role - Assigned role (e.g., TL, SM, GD, ADMIN)
- Email - User login ID
- Access - Access level identifier
- Active Status - Indicates whether the user account is active
- Created Date - Date and time of user creation
- Last Login - Most recent login timestamp

Search Functionality

The search bar allows administrators to search users by:

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- Name
- Email
- Role

This enables quick filtering of user records

User Actions

The Actions column provides administrative controls:

- Change Password - Reset or update the user's password
- Delete User - Remove a user from the system

3.3 Reset User Password

The Admin user can reset or change the password of any registered user through the User Management section. In the Manage User Login page, the administrator can view the list of users and their details. To change a password, click the Change Password icon in the

Actions column for the selected user. A Change Password dialog box will appear where the administrator can enter a new password and click Save to update it or Cancel to discard the changes.

Figure 4: Change Password Dialog Page

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4 Management Dashboard Overview:

The Manager Dashboard Overview page provides a summary of project statistics and quick access to project management features.

4.1 Dashboard Components

4.1.1 Summary

At the top of the dashboard, statistical summary cards display:

- Total Projects
- CFD Data Count
- Wind Data Count
- Flight Data Count
- Aero Data Count
- Total Reports

These cards provide a quick overview of system activity and uploaded dataset volumes.

Projects Section

The Projects panel displays all available projects in a structured list format.

Each project entry includes:

- Project Name
- Status (Active/Inactive)
- Short Description
- Created Date
- Member Count
- View Project Button

Clicking the View Project button navigates to the selected project's detailed page.

Data Distribution and Reports Panel

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The right-side panel displays:

- Data Distribution Overview
- Reports Summary

If no data is available, a placeholder message is shown. Once datasets and reports are added, the panels update automatically.

Figure 5: Dashboard Overview Page

To open a project, click the View Project button corresponding to the desired project.

After

clicking View Project, the system navigates to the Project Overview page. This page provides

access to the project workspace where users can manage datasets and navigate to different

modules

Figure 6: Project Overview Page

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4.1.2 Create New Project

Figure 7: Add Project Page

To create a new project:

- Click the + New Project button.
- Enter the Project Name.
- Provide a Project Description.
- Add Members (search by name or email).
- Click Create.

The newly created project will appear in the Projects list.

4.1.3 Notification Panel

Figure 8: Notification Panel Page

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The Notification Panel allows users to receive updates about important activities performed

within the project, such as completion of visualization processing or other system events.

The notification icon is located at the top-right corner of the application header. When a new

event occurs, the icon displays a notification count badge indicating the number of unread notifications.

By clicking on the notification icon, users can view a list of recent notifications along with the

corresponding message and timestamp. Notifications inform users about events such as visualization completion, system updates, or other processing results. Users can also mark

all notifications as read by selecting the "Mark all as read" option available in the notification panel.

4.2 Minutes of Meeting

The Minutes of Meeting module allows users to upload, manage, and track meeting minutes along with associated action items. Meetings can be organized under different meeting categories and linked to specific projects where applicable.

Figure 9: Minutes of meeting Page

4.2.1 Meeting Categories

Users can switch between different meeting types using the tabs:

- Technology Council (TCM)

- PMRC

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- Executive Board Meeting
- Group Director Meeting

For PMRC and Executive Board meetings, users have the option to associate the meeting minutes with a specific project.

4.2.2 Upload Meeting Minutes

Figure 10: Upload Meeting Minutes Page

To upload a new meeting record:

- Click Upload Minutes.
- Browse and select a PDF or Word file.
- Enter Action Points.
- Assign action owners (optional).
- Select the Meeting Date.
- Enter a Tag Name.
- Click Upload MOM.

Supported formats include PDF and Word documents.

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Managing Action Points

Each meeting can contain multiple action points assigned to users.

Users can:

- Add new action points
- Assign responsibilities to individuals or teams
- Edit existing actions
- Mark actions as completed

4.2.3 My Action - View and Edit

The "My Action" window allows users to view or modify assigned action points.

The page displays:

- Tag Name
- Meeting Date
- Assigned Person
- Action Description
- Completion Status

Users can switch between:

- View Action - Displays action details
- Edit Action - Modify action description or assignment

Actions can be marked as completed when tasks are finished.

Figure 11: View Action Page

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Figure 12: Edit Action Page

4.2.4 Task Assignment and Notification

When action points are assigned to users while creating or editing a meeting record, the system automatically sends a task notification to the assigned users.

Figure 13: Task Assigned Notification

The notification appears in the notification panel (bell icon) located at the top-right corner of the application.

The notification contains:

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- Meeting category
- Meeting title or tag name
- Meeting date
- Assigned action description

This alert informs the assigned user that a new task has been created and requires attention.

Users can open the notification and click Okay to acknowledge the assigned task.

4.2.5 Task Acknowledgement

After receiving the task notification, the assigned user can acknowledge the task by confirming the notification.

Figure 14: Task Acknowledged Notification

When the user clicks Okay, the system records the acknowledgement and sends a confirmation notification to the meeting creator.

The acknowledgement message informs the creator that the assigned user has reviewed and accepted the task.

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4.3 Student Engagement

The Student Engagement module allows users to manage student internship and academic programs. It provides functionality to add, edit, view, filter, and delete student program

records while tracking program progress and associated documentation.

Figure 15: Student Engagement Dashboard Page

4.3.1 Dashboard Overview

The top section displays summary cards such as:

- Total Students
- Internships
- Ongoing Programs
- Completed Programs

These cards provide a quick overview of student engagement statistics.

Users can also:

- Filter records by Type
- Filter records by Status

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Add new student program using the Add Student button

Student Programs Table

The Student Programs table displays all registered student records with the following details:

- Student Name
- College Name
- Project Title
- Program Type
- Duration
- Start Date
- End Date
- Guide / Mentor
- Status (Ongoing / Completed)
- Actions (View, Edit, Delete)

Users can perform the following actions:

- View program details

- Edit existing program information
- Delete a program record

4.3.2 Upload Student Program

To add a new student program:

- Click Add Student.
- Upload supporting document (PDF/Word format).
- Enter Student Name.
- Enter College Name.
- Enter Project Title.
- Select Program Type (e.g., Internship).

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- Select Start Date and End Date.
- Enter Program Duration (in months).
- Enter Guide Name.
- Select Program Status (Ongoing / Completed).
- Click Add Program to save.

Figure 16: Student Program Modal Page

4.4 Procurement Records

The Procurement Records module enables users to manage supply orders, track procurement expenses, allocate items to individuals or teams, and monitor order status.

4.4.1 Dashboard Summary

At the top of the page, summary cards display key procurement statistics:

- Total BO (Number of supply orders)
- Total Cost
- Ongoing Orders
- Completed Orders

These summary indicators provide a quick overview of procurement activity.

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Figure 17: Procurement Record Dashboard Page

Supply Order Table

The Supply Order table displays all procurement entries with the following details:

- ID
- Particulars
- Budget Name
- Quantity (QTY)
- Duration
- Start Date
- Delivery Date
- BO Order Number
- Purchase Order Number
- Amount
- Status
- Actions

Users can search records using the search bar and apply filters based on type or status.

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Status Tracking

Each procurement record displays its current status:

- Ongoing
- Completed

Status can be updated when delivery is completed or procurement is finalized.

Available Actions

In the Actions column, users can:

- View details
- Edit procurement record
- Delete record

4.4.2 Upload Supply Order

Figure 18: Add Supply Order Page

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1. To create a new procurement record:
2. Click the Procurement Record button.
3. Upload a supporting document (if required).
4. Enter the following details:

- BO Number
- Particulars
- Supply Name
- Quantity
- Delivery Date
- Start Date
- BO Order
- Purchase Order Number
- Status
- Quantity Allocation
- Amount

5. Click Submit Record to save.

4.4.3 Assigning Quantities

The system allows splitting the total quantity among different individuals, teams, or locations.

To assign quantities:

1. Click Manage Assignees.
2. Enter:
 - Number of Items
 - Assign To (Name or Location)
3. Click Add Assignee.

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4. Review:
 - Total Quantity
 - Assigned Quantity
 - Remaining Quantity
5. Click Done to confirm.

The system automatically calculates remaining quantity and prevents over-allocation.

Figure 19: Assignees Allocation Page

4.5 Budget Estimation

The Budget Estimation module allows users to create, manage, and track forecast budget records for different financial years. Users can upload supporting documents, enter

f

inancial details, and categorize expenses based on project requirements.

Figure 20: Budget Estimation Dashboard Page

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4.5.1 Budget Overview

The top section of the page includes:

- Search bar
- Financial Year selector
- Type filter
- Additional filter options
- Upload Forecast Budget button

These controls allow users to filter and manage budget records efficiently.

Forecast Budget Table

The Forecast Budget table displays all recorded budget entries with details such as:

- Cash Outgo Split Year
- TDCC Feasibility
- Reusability for Other Projects
- Necessity of Hardware Technologies
- Initiation of Condemnation of Store
- Capital or Revenue Classification
- Notes
- Attachment
- Action options

Users can:

- View attachment
- Edit budget record
- Download document
- Delete record

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Document Attachment

Each budget record can include a document attachment for reference. Users can:

- View attached file
- Download file
- Replace file during edit
- Supported formats: PDF and Word.

Editing and Deleting Records

Users can:

- Edit financial details
- Modify classification fields
- Update notes
- Delete records if no longer required

4.5.2 Upload Forecast Budget

To add a new forecast budget:

1. Click Upload Forecast Budget.
2. Upload supporting document (PDF/Word format).
3. Select Forecast Year.
4. Enter Division Name.
5. Enter Item details.
6. Enter Quantity (QTY).
7. Enter Existing Stock.
8. Enter Estimated Cost.
9. Enter Demand Indication Months.
10. Select Capital or Revenue.

11. Enter Project or Build details.
12. Enter DPP 2020 Serial Number.
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13. Enter Likely Cash Outgo amount.
14. Select TDCC feasibility.
15. Select Reusability for other projects.
16. Select necessity of hardware technologies.
17. Select initiation of condemnation of store.
18. Add Notes (if required).
19. Click Upload to save.

Figure 21: Upload Forecast Budget Page

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4.6 Divisional Records

The Divisional Records module allows users to manage and maintain departmental records with document attachments. Users can upload, view, edit, filter, download, and delete records based on type and status.

Figure 22: Division Record Dashboard Page

4.6.1 Record Distribution Overview

At the top of the page, summary cards display:

- Total Records
- Budget Records
- AMC Records
- Other Records

These cards provide a quick overview of record distribution by category.

Filtering Options

Users can filter records using:

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- Filter by Type (All Types, Budget, AMC, etc.)
- Filter by Status (All Status, Active, etc.)

This helps in quickly locating specific divisional entries.

Divisional Records Table

The table displays the following details

- Record Name
- Type
- Created Date
- Note
- Action

Under the Action column, users can:

- View record details
- Edit record information
- Download attachment
- Delete record

All updates reflect immediately after saving changes.

4.6.2 Upload Divisional Record

To add a new divisional record:

- Click Upload Record.
- Upload a document (PDF/Word format).
- Enter Division Name.
- Select Type (Budget / AMC / Others).

- Select Created Date.
- Enter Rating (if applicable).
- Add Note (optional).

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- Click Upload to save.

Figure 23: Upload Division Record Page

4.7 Customer Feedbacks

The Customer Feedbacks module allows users to record, manage, and review feedback submitted for various projects and divisions. The module supports ratings, document attachments, filtering, and complete feedback lifecycle management.

Figure 24: Customer Feedback Page

4.7.1 Feedback Overview Dashboard

The top section displays summary cards including:

- Total Feedbacks
- Average Rating

These provide a quick performance overview across divisions.

Filtering Feedback

Users can filter feedback records using:

- Filter by Division dropdown

This allows users to narrow down results based on specific divisions.

Feedback Records Table

The table displays:

- Project Name
- Division
- Note
- Ratings (displayed with star indicators)
- Feedback Date
- Action

Under the Action column, users can:

- View feedback details
- Edit feedback
- Download attached document
- Delete feedback

All updates are reflected immediately after saving changes.

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4.7.2 Upload / Edit Feedback

To add or modify feedback:

- Click Upload Feedback.
- Upload a supporting document (PDF/Word format).
- Enter Project Name.
- Enter Division.
- Provide Rating.
- Select Feedback Date.
- Add Notes (optional).

- Click Save.

Figure 25: Upload Document Page

4.8 Training Records

The Training Records module allows users to manage and track employee training activities.

It supports uploading training documents, maintaining training status, and performing full

record lifecycle operations including view, edit, download, and deletion.

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Figure 26: Training record Document Page

Overview Section

At the top of the page, summary cards display:

- Total Trainings
- Number of Participants
- Ongoing Trainings

These cards provide a quick overview of training activity status.

Search and Filters

Users can manage records using:

- Search bar (Search training name or trainee)
- Filter by Type (e.g., Pre-Training, Post-Training)
- Filter by Status (Ongoing, Completed)

Training Records Table

The table includes:

- Name
- Training Name
- Type

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- Start Date
- End Date
- Status
- Remarks
- Action

Under the Action column, users can:

- View training details
- Edit training record
- Download attachment
- Delete record

Status indicators clearly show whether a training is Ongoing or Completed.

Deletion and Confirmation

Before deleting a training record, the system prompts a confirmation message to prevent accidental data loss.

4.8.1 Upload / Edit Training Record

To add or update a training record:

- Click Upload Training.
- Upload supporting document (PDF/Word format).
- Enter Trainee Name.
- Enter Training Name.
- Select Training Type.
- Enter Start Date.

- Enter End Date.
- Select Status.
- Add Notes (optional).
- Click Upload to save.

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Figure 27: Upload Document Page

4.9 Technical & Design Reports

Figure 28: Technical and Design Report Dashboard Page

The Technical & Design Reports module allows users to manage technical, design, and other

categorized reports. Users can upload documents, classify reports by type, search records,

and perform complete lifecycle operations including view, edit, download, and deletion.

4.9.1 Overview Section

At the top of the page, summary cards display:

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- Total Records
- Technical Reports Count
- Design Reports Count

These cards provide a quick statistical overview of stored reports.

Search and Filter Options

Users can locate specific reports using:

- Search bar (Search reports, tags, projects)
- Filter by Type (Technical, Design, Other)

This improves navigation and record accessibility.

Reports Table

The table displays:

- Report Name
- Type
- Created Date
- Note
- Action

Under the Action column, users can:

- View report details
- Edit report metadata
- Download attached document
- Delete report

Changes are reflected immediately after saving.

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4.9.2 Upload / Edit Report

Figure 29: Upload Document Page

To add or modify a report:

- Click Upload Document.
- Upload a document (PDF/Word format).
- Enter Report Name.
- Select Report Type (Technical / Design / Other).
- Select Created Date.
- Add Note (optional).
- Click Upload to save.

4.10 Digital Library

The Digital Library module allows users to store, manage, and organize uploaded documents. Users can upload files, search records, apply filters, sort results, and perform

file operations such as view, edit, download, and delete.

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Figure 30: Digital Library dashboard Page

4.10.1 Search and Filtering

Users can manage files using:

- Search bar (Search by file name, tags, or keywords)
- Filter by Type
- Sort option (Newest, etc.)

These options allow quick retrieval and organization of stored documents.

Files Table

The table displays:

- File Name
- Type
- Size
- Created Date
- Action

Under the Action column, users can:

- View file
- Edit file metadata

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- Download file
- Delete file

All changes are reflected immediately after saving.

4.10.2 Upload File

To upload a new file:

- Click Upload File.
- Select document (PDF/Word format).
- Click Upload to save the file.

Figure 31: Upload Document Page

14.11 Settings

The Settings module allows users to manage their account credentials and view system related information, including application details and version information.

Figure 32: Settings Page

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4.11.1 Reset Password

The Reset Password section allows users to securely change their account password.

Users must:

- Enter Current Password.
- Enter New Password.
- Re-enter New Password for confirmation.
- Click Save Password.

Password requirements:

- Minimum 8 characters
- At least one special character

If the entered passwords do not match or do not meet requirements, validation messages are displayed. The system ensures secure password handling.

About the Tool

The About section provides a brief description of the application, explaining its purpose and functionality. It highlights:

- Data analysis capabilities
- Visualization features
- Interactive reporting support

This section helps users understand the purpose and scope of the system.

Version Information

The version section displays the current application version (e.g., Version 2025.0).

This helps in:

Tracking system updates

- Identifying build versions
- Supporting maintenance and troubleshooting

5 Project Overview

At the top of the page, basic project information is displayed:

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- Project Name
- Project Status (e.g., Active)
- Project Description
- Created Date
- Member Count

Figure 33: Project Overview Page

5.1 Dataset Tabs

Project data is categorized into four dataset types:

- CFD Data
- Wind Tunnel Data
- Flight Data
- Others

Key behavior:

- Only one dataset tab can be active at a time.
- Switching between dataset tabs refreshes the displayed tags and file list.
- Each dataset maintains its own independent set of tags and associated files.

This ensures structured organization of project data by category.

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Tag / Folder Management

Within each dataset, files are organized under Tags or Folders.

The table displays:

- Tag / Folder Name
- Processing Status (if applicable)
- Created Date
- Action options (Edit, Delete)
- Go To navigation button

Users can:

- Rename a tag
- Delete a tag
- Navigate to tag-specific data

This section provides a quick summary of the selected project.

5.2 Edit Project

Users can edit project details using the Edit option.

The Edit Project modal allows:

- Viewing the Project Name (cannot be changed after creation).
- Updating the Project Description.
- Adding new members using name or email search.
- Removing existing members.
- Saving changes or cancelling the update.

The project name remains fixed to maintain data integrity.

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Figure 34: Project Edit Page

5.3 Project Members

The Project Members view displays:

- Member Name
- Member Email ID

This provides a clear list of all users assigned to the project.

Figure 35: Project Members Page

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5.4 Upload File

Figure 36: Upload Files Page

Clicking the Upload File button opens the upload modal.

The upload modal allows users to:

- Browse and select plot files.
- Preview selected files (if supported).
- Enter Folder / Tag Name.
- Select Data Type (e.g., CFD).
- Upload selected files.

Upload Modal Modes

The upload modal operates in two modes:

Create Mode

- Upload files under a new tag.
- A new folder is created for the selected dataset.

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Edit Mode

- Rename an existing tag.
- Upload additional files to an existing tag.

File Processing and Navigation

After upload:

- Files are associated with the selected dataset and tag.
- Users can navigate to the dataset view using the "Go To" action button.
- Processing status indicators (if applicable) are displayed next to tags.

File Selection

Select Files

- The user clicks Browse Plot Files to select one or more files.
- Multiple files can be selected at once.

Supported File Types

File behavior depends on the format:

1. CSV / Excel / TXT / DAT / C files

- Eligible for preview.
- Can be used for visualization.

2. Images and other file formats

- Stored as raw files only.

After selection, the system displays:

- File Name
- File Type
- File Size
- Visualization eligibility (enabled only for supported tabular files)

Provide tag and data type

- the user enters a Folder / Tag Name, which groups the uploaded files.
- The user selects a Data Type (CFD, Wind Data, Flight Data. Others). In edit mode, the data type is fixed and cannot be changed.

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5.5 Configure Plot File Headers

Once tabular file is selected, the Plot File Header option appears.

Users can choose how headers are handled:

- Use headers from file - system reads the first row
- File has no headers - system auto-generates column names.
- Provide custom headers - user manually enters comma-separated headers.
- When custom headers are used: users can manually define custom headers, which can be applied to one or multiple selected files using the "Apply custom headers to all files" option

Figure 37: Upload Files Page

5.6 Select file for preview

? Clicking a file from the list loads it into the Preview panel.

? The preview adapts automatically based on file type.

Image files: a visual preview of the image is displayed.

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Figure 38: Image Files Upload and preview Page

Excel files: all available sheets are listed. Users can switch between sheets and select which sheets should be included for visualization.

Figure 39: Upload and preview of Excel Files

Dat / .c text files:

1. The preview displays the line-by-line with line numbers.
2. Total line count is shown.
3. The user can select start line and end line.

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4. Only the selected range is parsed, used for data extraction, used for plotting graphs.

5. Selected range is visually highlighted.

6. The system auto-detects headers from the first selected line.

7. The parsed data range is also shown as a table preview.

Figure 40: Upload and preview for .Dat Files

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- Apply line range to multiple files

Figure 41: Apply range for Files Page

? For .Dat and .c files, users can enable "Apply range to all .Dat / .c files".

? This applies the same start and end line selection to all similar files, ensuing

consistency and reducing manual effort.

MAT (MATLAB) Files:

? The preview displays all variables available in the .mat file along with their dimensions and data types.

? When a variable is selected, its data is shown in a tabular format.

? Users can define a slice range to load only a specific portion of the data.

? After clicking Apply, only the selected slice is displayed and processed.

? The selected data is prepared for further parsing and visualization.

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Figure 42: MAT file preview

- Upload and processing

? When user clicks upload.

? A progress bar shows upload status.

? Files are uploaded along with: dataset type, tag name, header configuration, selected sheets, selected line ranges.

? Visualization-enabled files are queued for processing.

? Non- visualized files are stored as raw files.

Figure 43: Stored Files Page

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5.7 Raw preview page:

The Raw Preview page displays the data exactly as it was uploaded, without any changes.

Figure 44: Raw Preview Page

5.8 Processed Preview Page:

The Processed Preview page displays data that has been parsed and prepared for plotting. Users can edit the column headers on this page.

Figure 45: Processed Preview Page

5.9 Global Search

The Global Search feature allows users to quickly find different sections, tags, files, and visualizations within a project. The search bar is located at the top-right corner of the Project Overview page.

Users can enter keywords related to project sections or content. As the user types, the system displays matching results such as Project Overview, Visualize, Minutes of the Meeting, Technical Reports, Digital Library, Student Engagement, and other available sections within the project. Selecting a result will automatically navigate the user to the corresponding section.

This feature helps users quickly access specific project components without manually browsing through the navigation menu.

Figure 46: Global Search Page

6 Visualization:

The Visualization module enables users to generate interactive analytical plots from processed datasets across CFD, Wind Tunnel, and Flight data domains.

The system supports:

- Multi-dimensional plotting (2D & 3D)
- Large dataset visualization

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- Multi-series comparison (Overplot)
- Logarithmic scaling
- Derived data visualization

6.1 Visualization Workflow

Dataset

?

Tag / Folder

?

Processed File

?

Column Selection

?

Plot Type Selection

?

Axis Assignment

?

Generate Plot

?

Preview / Save

6.2 Plot Type Classification

A. 2D Plots (Primary Analysis)

- Line Plot
- Scatter Plot
- Bar Chart
- Histogram
- Box Plot
- Violin Plot

Used for trend analysis, comparisons, and distributions.

B. Specialized 2D Plots

- Polar Plot

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- Contour Plot
- Heatmap

Used for aerodynamic and multi-variable analysis.

C. 3D Plots

- 3D Scatter
- 3D Line
- 3D Surface

Used for multi-dimensional visualization.

6.3 Plot Descriptions

Line Plot

Represents trends over continuous variables.

Example: Altitude vs Time (Flight Data)

Scatter Plot

Shows relationship between two variables.

Example: Lift vs Drag

Bar Chart

Compares discrete categories.

Histogram

Shows frequency distribution.

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Box / Violin

Displays statistical spread and density.

Contour Plot

Requires X, Y, Z data

Displays value distribution as contour levels

Example: CL vs Mach & Angle of Attack

Polar Plot

Used for angular datasets

Example: Lift coefficient vs Angle of Attack

Heatmap

Displays intensity variation

3D Surface Plot

Represents continuous surface

Example: Pressure distribution over wing surface

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6.4 2D Chart Type

Figure 47: 2D Chart Type

2D Chart Visualization

Figure 48: 2D Linear Chart Visualization Page

The line chart connects data points sequentially to represent changes in the dataset over the selected axis values. This type of visualization is useful for analyzing patterns, variations, and trends in continuous data.

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Figure 49: 2D Bar Chart Visualization Page

Bar charts are useful for comparing values across different categories or intervals in the dataset. Users can easily identify variations, differences, and distribution of data through the bar representation.

6.5 3D Chart Type

Figure 50: 3D Chart Type

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- 3D Chart Visualization

Figure 51: 3D Chart Visualization Page

This figure illustrates a 3D visualization where three parameters are plotted along the X, Y, and Z axes. Users can rotate and inspect the plot interactively.

- Assign Axes

? Once a file is selected, available columns from the file are loaded.

? X-axis and Y-axis scales (Linear or Log) can also be configured.

? The user assigns:

? X-axis and Y-axis for all plots

? Z-axis, if the selected chart type requires it (e.g., 3D or contour plots)

6.6 Cross Plotting

The system provides a Cross Plotting feature that allows users to inspect dataset values before selecting X-axis and Y-axis parameters.

When users click the eye icon next to the X-axis or Y-axis field, a Data Browser window is opened.

This Data Browser provides a tabular view of the selected dataset file.

Steps:

- Select Dataset, Tag, and File.

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- Click the eye icon next to:

? X Axis OR

? Y Axis

- The Data Browser window will open.

The table displays:

? Row numbers

? Column names (column1, column2, etc.)

? Corresponding data values

- Users can:

? Scroll through the dataset

? Inspect values in each column

? Identify appropriate columns for plotting

- After reviewing the data:

? Close the Data Browser

? Select the required column from the dropdown

Figure 52: 3D Cross Plotting

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Figure 53: Cross Plotting with preview Option

- Plot name (Optional)

? User can provide a custom plot name.

? If no custom name given, the system automatically generates a meaningful name based on the dataset and axis selection.

Figure 54: Plot Setting

- Generate Visualization

? After configuration, the user clicks Generate plot.

? The system submits the selected configuration and starts processing file visualization.

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Figure 55: Visualization Page

6.7 Axis Configuration (Log)

- In the visualization settings, users can configure the Log scale for the X-axis and Y axis to represent data on a logarithmic scale.

- The Log scale is useful when the dataset values span across multiple orders of magnitude, allowing better visualization of variations in the data.

- When the Log option is selected in the X Scale and Y Scale dropdown menus, the axis values are displayed in logarithmic format instead of a standard linear progression

- When the Log option is selected for either the X Scale or Y Scale, the chart is displayed in semi-log format, where one axis uses a logarithmic scale while the other axis remains in standard linear scale. This helps visualize data that spans multiple orders of magnitude along one axis.

- X-axis and Y-axis required for all plots

- Z-axis required for:

- Contour

- 3D plots

Scale Options:

- Linear
- Logarithmic

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- Semi-log

Figure 56: Log Scale Selection in Axis Configuration

Figure 57: Visualization with Logarithmic Axis Scale

Figure 58: Visualization with semi-log Axis Scale

6.8 Large Dataset Handling

To support large datasets:

- Chunk-based processing
- Progressive rendering
- Data preview limits
- Tile-based loading (LOD)

Benefits:

- Faster performance
- Smooth zooming and interaction

6.9 MAT File Visualization

The system provides a dedicated interface for visualizing MATLAB (.mat) files. Unlike standard datasets, .mat files contain variables that must be selected before generating plots.

MAT File Selection

In the Visualization module:

- Select Dataset
- Select Tag (MAT_FILE)
- Select the required .mat file from the File dropdown
- After selecting the .mat file, the system automatically loads available variables.

Plot Configuration (MAT Plot Setting)

The Plot Setting section allows users to configure visualization for .mat files.

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Users can:

- Select MAT Chart Type (e.g., Scatter)
- Choose Signature:
 - Plot(X)
 - Plot(X, Y)

Plot Builder

The Plot Builder is used to select variables from the .mat file.

Users must:

- Select X Variable
- Select Y Variable

Each variable displays:

- Variable name

- Dimensions

Slice Selection

To handle large datasets, slicing is available:

- Enter slice values
- Example: :, 0:10
- Click Apply (if required)
- The system will use only the selected portion of data.
- This improves performance when working with large arrays.

Generating Visualization

- Click Generate Plot

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- The system processes selected variables and generates the visualization.
- The generated plot appears in the lower section of the screen.

Users can:

- View the plotted data
- Verify the selected variables
- Save the visualization

Figure 59: MAT File Plot Generation

6.10 Create multiple plot (Over-plot)

? Overplot allows comparison of multiple datasets simultaneously

? Supports:

- Same file multiple columns
- Multiple files
- Cross-dataset comparison

? Ensures:

- Consistent axis scaling
- Clear legends

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- Toggle visibility

? Users can add multiple plot series to overlay data on the same chart.

? Each series can:

- ? Use a different dataset, file, or axis configuration.
- ? Be individually enabled or disabled
- ? Removeable if not required

? Users can overlay multiple datasets on the same plot.

? Capabilities:

- Compare multiple files
- Compare across datasets (CFD vs Flight)
- Enable/disable individual series
- Shared axis configuration

Figure 60: Overplot Visualization Page

- Preview Visualization

? While processing, the system displays status updates.

? Once ready, the generated plot is shown in an interactive preview area.

? Users can zoom, pan, and inspect the plot directly in the preview.

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? Users can:

- ? Zoom
- ? Pan

? Hover for values

? Toggle series

Figure 61: Preview Visualization Page

- View data tiles (optional)

? For large datasets, the system allows users to preview subsets (tiles) of the underlying data.

? This helps users validate the plots are automatically saved.

- Save and manage visualizations

? Saved visualization are listed on the same page, where users can:

? Reload a visualization.

? Open in full-screen view

? Delete it if no longer needed

Figure 62: Saved visualization files Page

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7 Calculation Module

The Calculation module allows users to compute derived parameters from the uploaded datasets. Users select the Dataset, Tag, and File, then enter the required mathematical formula and map the variables to the corresponding dataset columns. After clicking Process Formula, the system calculates the result and displays a preview of the computed values. Once the results are verified, the derived column can be saved for further analysis or visualization.

Figure 63: Calculation Page

Figure 64: Turn Radius Estimation

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s

Figure 65: True Air Speed (TAS) from Mach

8 Aircraft Performance Module

The Aircraft Performance Module enables users to analyze aircraft performance characteristics using the available dataset. The system allows users to generate different

performance plots based on the selected parameters from the input dataset.

Using the available aircraft performance data, users can generate plots to analyze aircraft

operational limits, climb performance, aerodynamic characteristics, and maneuvering behavior. These plots help in understanding how the aircraft performs under different flight

conditions such as varying speed, altitude, and load factors.

Figure 66: Rate of Climb vs Mach Number Plot

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Figure 67: Aircraft Turn Rate Performance Plot